

Systematic Comparative Study of Retrieval Methods for Construction Catalogs

Lexical retrieval achieves **94% Acc@1** in 40K+ item catalog search

CONTEXT

40K+

Catalog Items

"2 tubes PVC 110 mm"

EXACT MATCH?

Small errors = Costly mistakes

METHOD

16,590

Test Queries

LEXICAL: TF-IDF • BM25

NEURAL: E5 • GTE • BGE

HYBRID: RM3 • Cross-enc

RESULTS

BM25 **97.4%** ✓

Neural **44.8%**

Hybrid **89%**



97.4% Accuracy

Open Dataset
~40K items

Open-source
Framework

Automation in Construction

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González-Alvarez et al. (2025)
Automation in Construction